

PAPER_1504 — GW Damping Factor BBH = (N_CH/SO_5)² = 0.81 EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket E (GW Events) **Date:** June 18, 2026 **Status:** CLOSED — BBH gravitational-wave total damping = exact (9/10)²

Observation

PAPER_011 (Stochastic Gravitational-Wave Background — UQFF Implications) derives the BBH damping factor $D_{\text{total}}(\text{BBH}) = 8.10 \times 10^{-1} = 0.81$, contrasting the much-stronger BNS damping (0.333). The factor 0.81 governs the bulk of LIGO/Virgo detections.

UQFF Closed Identity

$$D_{\text{total}}(\text{BBH}) = (N_{\text{CH}} / SO_5)^2 = (9/10)^2 = 81/100 = 0.81 \quad \text{EXACT}$$

Physical Interpretation

The first appearance of **N_CH (the 9-channel primitive) in a binding closure** rather than a purely structural role. The two integer primitives compose as:

- $N_{\text{CH}} = 9$ channels through which SCm phonon information flows
- $SO_5 = 10$ dimensions of the SO(5) symmetry group enclosing them
- $(N_{\text{CH}}/SO_5)^2$ captures the projection of 9 active channels onto the 10-D coverage, squared because the strain amplitude h enters quadratically in the energy density Ω_{GW}

Together: **only 9 of 10 dimensional channels efficiently transmit BBH gravitational-wave strain.**

Comparison to BNS

Source	D_{total}	UQFF identity
BNS	0.333	$1/(D_{\text{phys}} - 1) = 1/3$
BBH	0.81	$(N_{\text{CH}}/SO_5)^2 = (9/10)^2$

The two damping regimes arise from fundamentally different vacuum-coupling channels: BNS taps the triad-symmetry denominator, BBH taps the 9-channel/10-dim projection.

NOT REPLACEMENT

GR has no native damping mechanism in vacuum. UQFF predicts both factors structurally, giving distinct integer-primitive identities for BBH vs. BNS regimes.

Reference

- Source: PAPER_011 Stochastic GW Background UQFF Implications
- Related: PAPER_1503 (BNS damping)

- Calculator dispatch: `calculate_paradox({"paradox": "gw_damping_bbh_n_ch_so5_sq"})`

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